

Certificate status

Repository source: `certificate/STATUS.md`

Claim

The exact package represents the proposed universal theorem

$$\text{Vol}(K) \geq \frac{26221074253\pi}{200000000000} d^3 = 0.41187967121228637754323153860 \dots d^3$$

for every three-dimensional convex body of constant width d .

Current classification

PROPOSED EXACT UNIVERSAL CERTIFICATE - STANDALONE RELEASE v1.1.0

The public-format package is independent of the private repository layout and replays using only files contained in this fresh-history repository. Its exact Python verifier, independent Python binary64 audit, independent base-R audit, seven adversarial mutations, paper/document checks, hosted three-job workflow, and named-author fresh-clone replay pass in the recorded environments.

The audited pre-release state is `892d4bc6fad07950e78da66e45b95063a6415af6`. Its successful hosted workflow is <https://github.com/swihart/minvol-degree3-contact-bound/actions/runs/35468019663>. The release metadata is `v1.1.0`, dated September 26, 2026. The immutable tag and GitHub Release remain operational publication gates until their workflows and downloaded assets are verified.

Established inside this package

- gap-free circumradius coverage from $1/2$ to $\sqrt{6}/4$;
- lower-branch weakest row `03B02B`; the universal bottleneck is the near-Jung branch;
- four exact replacement cells for the old `03A1` interval;
- dominant-edge coordinate-width sharpening on all three axes;
- exact final-fan reconstruction and fan-margin verification;
- determinant transfer and forced-core lower bound;
- three exact cap-pair lower bounds and six-cap disjointness;
- strict exact near-Jung handoff surplus;
- exact theorem coefficient and decimal enclosure;
- byte-identical reconstruction of the authoritative certificate JSON;
- independent base-R reconstruction on both the named-author machine and hosted R 4.6.1;
- rejection of targeted lower-row, fan-radius, multiplier, section-hull, cap-allocation, disjointness, and theorem-coefficient mutations;
- automatic reconciliation of the focused Paper v1 theorem constants and lower-branch table against the authoritative certificate JSON;
- generated exact-constants and release-document consistency checks;
- paired Markdown/PDF renderings of the complete release-document set;
- successful hosted Poppler preflight for committed and rebuilt PDFs; and
- a complete clean replay from a fresh clone of the staging remote.

The hosted and fresh-clone evidence is indexed in [evidence/rc1_remote_audit.json](#).

Not established

- external independent mathematical reproduction;
- peer review or journal acceptance;
- sharpness, equality, or rigidity;
- identification of a minimizer;
- Meissner extremality;
- an archival DOI or external subject-matter reproduction.

Remaining publication gates

1. green hosted Python, R, and document jobs on the final release commit;
2. an annotated `v1.1.0` tag pointing to that exact commit;
3. clean tagged-checkout replay and green hosted jobs on the tag;
4. tagged release assets and an independently checked download manifest;
5. explicit named-author approval before changing visibility, creating the GitHub Release, or announcing the result; and
6. external mathematical review or reproduction when available.

Authorship, no-affiliation wording, citation metadata, repository identity, release date, version, and the split CC BY 4.0/MIT license are fixed for `v1.1.0`.